Light

GCSE Computer Science - CPU Architecture (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main function of the Arithmetic Logic Unit (ALU) in a CPU?

- ○ **A.** To store the instructions that are currently being executed
- ○ **B.** To coordinate and control all operations within the CPU
- ○ **C.** To perform calculations and logical comparisons
- ○ **D.** To generate timing signals for the processor

Q2 (1 mark)

Which CPU component is responsible for decoding instructions and coordinating the activities of other components?

- ○ **A.** The control unit

- ○ **B.** The arithmetic logic unit
- ○ **C.** The system clock
- ○ **D.** The data bus

Q3 (1 mark)

What is the purpose of the clock in a CPU?

- ○ **A.** To display the current time on the computer screen
- ○ **B.** To synchronise the operations of CPU components by generating regular timing signals
- ○ **C.** To store the time when files were last modified
- ○ **D.** To control the brightness of the monitor

Q4 (1 mark)

What is a register in the context of CPU architecture?

- ○ **A.** A type of secondary storage device
- ○ **B.** A connection between the CPU and RAM
- ○ **C.** Software that records user activity
- ○ **D.** A small, fast storage location within the CPU that holds data temporarily during processing

Q5 (1 mark)

In computer architecture, what is a bus?

- ○ **A.** A collection of wires that transmit data and signals between components
- ○ **B.** A type of processor designed for graphics
- ○ **C.** Software that manages hardware devices
- ○ **D.** A cooling system for the CPU

Q6 (1 mark)

Clock speed is typically measured in which unit?

- ○ **A.** Bytes per second
- ○ **B.** Megapixels
- ○ **C.** Gigahertz (GHz)
- ○ **D.** Watts

Q7 (1 mark)

What does it mean when a CPU is described as 'quad-core'?

- ○ **A.** It has four times the normal clock speed
- ○ **B.** It contains four separate processing units that can work simultaneously

- ○ **C.** It can connect to four different devices at once
- ○ **D.** It uses four different types of memory

Q8 (1 mark)

What is the primary purpose of cache memory?

- ○ **A.** To provide permanent storage for the operating system
- ○ **B.** To connect the CPU to external devices
- ○ **C.** To back up important files automatically
- ○ **D.** To store frequently accessed data close to the CPU for faster retrieval

Q9 (1 mark)

Why does the CPU need to access main memory (RAM)?

- ○ **A.** To retrieve instructions and data for programs that are currently running
- ○ **B.** To permanently save files when the computer shuts down
- ○ **C.** To display images on the monitor
- ○ **D.** To connect to the internet

Q10 (1 mark)

Which of the following operations would be performed by the ALU?

- ○ **A.** Fetching an instruction from memory
- ○ **B.** Comparing two numbers to determine which is larger
- ○ **C.** Sending a signal to the printer
- ○ **D.** Storing a file on the hard drive

Q11 (1 mark)

What type of information does the address bus carry?

- ○ **A.** The actual data being processed
- ○ **B.** Control signals to coordinate operations
- ○ **C.** The memory locations where data should be read from or written to
- ○ **D.** Error messages from the CPU

Q12 (1 mark)

What does the control unit use to tell other components what to do?

- ○ **A.** Mathematical calculations
- ○ **B.** Sound frequencies
- ○ **C.** Stored user passwords
- ○ **D.** Control signals sent via the control bus

Q13 (2 marks)

A computer's CPU has its clock speed increased from 2.5 GHz to 3.5 GHz. What effect would this most likely have on performance?

- ○ **A.** More instructions can potentially be executed per second, improving processing speed for most tasks
- ○ **B.** The computer will use less electricity
- ○ **C.** The amount of RAM in the system will increase
- ○ **D.** Files will be permanently stored more quickly

Q14 (2 marks)

Why is cache memory faster than main memory (RAM)?

- ○ **A.** Cache has a larger storage capacity than RAM
- ○ **B.** Cache is connected to the internet while RAM is not
- ○ **C.** Cache stores data permanently while RAM is temporary
- ○ **D.** Cache is located closer to the CPU and uses faster memory technology

Q15 (2 marks)

A user upgrades from a dual-core processor to a quad-core processor with the same clock speed. In which situation would they see the greatest performance improvement?

- ○ **A.** Running a single program that performs one calculation at a time

- ○ **B.** Viewing a static image on the screen
- ○ **C.** Running multiple applications simultaneously, such as a web browser, music player, and word processor
- ○ **D.** Typing a document in a word processor

Q16 (2 marks)

During a calculation, how do registers and the ALU work together?

- ○ **A.** Registers perform the calculation and the ALU stores the result permanently
- ○ **B.** Registers hold the data values, the ALU performs the calculation, and the result is placed back into a register
- ○ **C.** Registers and the ALU are located in different computers and communicate via the internet
- ○ **D.** The ALU fetches data directly from the hard drive and registers display the result

Q17 (2 marks)

A computer system has three main buses: address bus, data bus, and control bus. Why is having all three types necessary?

- ○ **A.** Each bus type carries different information needed for the CPU to communicate effectively with memory and other components
- ○ **B.** Only one bus is actually needed; the others are backup systems
- ○ **C.** The buses compete with each other to see which can transfer data fastest

- ○ **D.** Each bus connects to a different type of secondary storage

Q18 (2 marks)

Many modern CPUs have multiple levels of cache (L1, L2, L3). Why is cache organised in this way?

- ○ **A.** Each level stores different types of files
- ○ **B.** The different levels provide a balance between speed and capacity-L1 is fastest but smallest, L3 is slower but larger
- ○ **C.** L1, L2, and L3 are different brands of cache made by different manufacturers
- ○ **D.** Only L1 is used by the CPU; L2 and L3 are for display graphics

Q19 (2 marks)

Why can't manufacturers simply keep increasing CPU clock speed indefinitely to improve performance?

- ○ **A.** Higher clock speeds generate more heat and consume more power, creating physical limitations
- ○ **B.** There is a legal limit on how fast CPUs can operate
- ○ **C.** Higher clock speeds require more RAM to be installed
- ○ **D.** Clock speed has no effect on performance anyway

Q20 (2 marks)

How does increasing the size of cache memory improve CPU performance?

- ○ **A.** Larger cache allows the CPU to run at higher clock speeds
- ○ **B.** Larger cache increases the number of cores in the CPU
- ○ **C.** Larger cache provides permanent storage for the operating system
- ○ **D.** Larger cache can hold more frequently used data, reducing the number of slower accesses to main memory

Q21 (2 marks)

Why do CPUs use registers to hold data during processing rather than accessing RAM directly for every operation?

- ○ **A.** Registers have unlimited storage capacity
- ○ **B.** RAM cannot store numbers, only text
- ○ **C.** Registers are located within the CPU and can be accessed in a single clock cycle, whereas RAM access takes many cycles
- ○ **D.** RAM is only used for displaying graphics

Q22 (2 marks)

A computer has a 64-bit data bus instead of a 32-bit data bus. How does this affect data transfer?

- ○ **A.** The data bus can now only transfer half as much data
- ○ **B.** Twice as much data can be transferred in a single operation, potentially improving performance for data-intensive tasks
- ○ **C.** The computer will use twice as much electricity

○ **D.** The clock speed will automatically double

Q23 (2 marks)

During program execution, the control unit sends a 'read' signal to memory via the control bus. What happens next in this process?

○ **A.** The memory deletes all stored data

○ **B.** The user must manually approve the read operation

○ **C.** The memory retrieves data from the location specified on the address bus and places it on the data bus

○ **D.** The ALU automatically performs a calculation

Q24 (3 marks)

A software company is selecting a CPU for a server that will run many simultaneous database queries from different users. The options are:

- CPU A: 4 cores at 4.0 GHz with 8 MB cache
- CPU B: 8 cores at 3.0 GHz with 16 MB cache

Which CPU would likely perform better for this workload, and why?

○ **A.** CPU A, because higher clock speed is always the most important factor

○ **B.** CPU A, because fewer cores are easier to program

○ **C.** CPU B, because more users means higher total storage is needed

○ **D.** CPU B, because handling many simultaneous queries benefits from more cores for parallel processing, and the larger cache helps with repeated data access

Q25 (3 marks)

A program contains the instruction to add two numbers stored in memory and save the result. Describe the correct sequence of component involvement.

- ○ **A.** The hard drive performs the calculation; RAM stores the result; the clock displays the answer on screen
- ○ **B.** ALU fetches the instruction; registers perform the addition; control unit stores the result; cache deletes the original numbers
- ○ **C.** Control unit fetches the instruction; control unit directs memory to send the numbers via the data bus; registers hold the numbers; ALU performs the addition; result is stored in a register then written to memory
- ○ **D.** Memory performs the addition directly; the result bypasses the CPU entirely

Q26 (3 marks)

A laptop manufacturer must choose between two CPU designs:

- Design X: Very high clock speed (5.0 GHz) but only 2 cores
- Design Y: Moderate clock speed (2.5 GHz) but 8 cores

Which design would be more appropriate for the laptop, considering that laptops must balance performance with battery life and heat generation?

- ○ **A.** Design X, because the laptop has unlimited battery power
- ○ **B.** Design Y, because it provides good multi-tasking performance while generating less heat and using less power than the extremely high-frequency Design X
- ○ **C.** Design X, because laptops never need to run multiple programs
- ○ **D.** Neither design would work in a laptop

Q27 (3 marks)

A CPU is running a program that processes a very large dataset that cannot fit entirely in cache. The program repeatedly accesses different parts of this dataset. What performance issue would occur, and why?

- ○ **A.** The monitor would stop displaying images
- ○ **B.** The program would run significantly slower due to frequent cache misses, requiring many time-consuming accesses to main memory instead of fast cache accesses
- ○ **C.** The CPU would become permanently damaged
- ○ **D.** The keyboard would stop responding

Q28 (3 marks)

A CPU has a 32-bit address bus. What is the maximum amount of memory this CPU can directly address, and why?

- ○ **A.** 32 bytes, because each bit represents one byte of storage
- ○ **B.** 32 megabytes, because the address bus width equals megabytes
- ○ **C.** Unlimited memory, because the address bus can be extended
- ○ **D.** 4 gigabytes (approximately 4.3 billion bytes), because 2^3^2 unique addresses can be represented, and each address refers to one byte

Q29 (3 marks)

A computer has a very fast CPU but relatively slow RAM. Programs frequently stall while waiting for data. What is this performance limitation called, and how might it be addressed?

○ **A.** This is called a memory bottleneck; it can be addressed by increasing cache size to reduce the frequency of slow RAM accesses, or by using faster RAM

○ **B.** This is called clock overflow; it requires reducing the CPU speed

○ **C.** This is called register exhaustion; it requires adding more hard drive space

○ **D.** This is called bus underflow; it requires removing RAM from the system

Q30 (3 marks)

Two CPUs are being compared:

- CPU P: 4 cores, 3.8 GHz, 12 MB cache, designed for high single-thread performance
- CPU Q: 16 cores, 2.4 GHz, 32 MB cache, designed for parallel workloads

For a video editing application that can utilise multiple cores for rendering but also requires fast response during preview playback, which CPU would provide the better overall experience?

○ **A.** CPU P only, because video editing never benefits from multiple cores

○ **B.** CPU Q provides better balance: 16 cores significantly accelerate rendering tasks, while 2.4 GHz is still adequate for preview playback, and the larger cache helps with large video files

○ **C.** Neither CPU is suitable for video editing

○ **D.** The CPUs would perform identically regardless of the application

Total: **55** marks • Time: **30 min**

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